

Appendix B: The detailed methodology for developing the design principles

The diagrams of station areas

As shown in the following diagrams, and the on-site visits, the station areas are hardly perceivable (by vision) outside the 250m range.

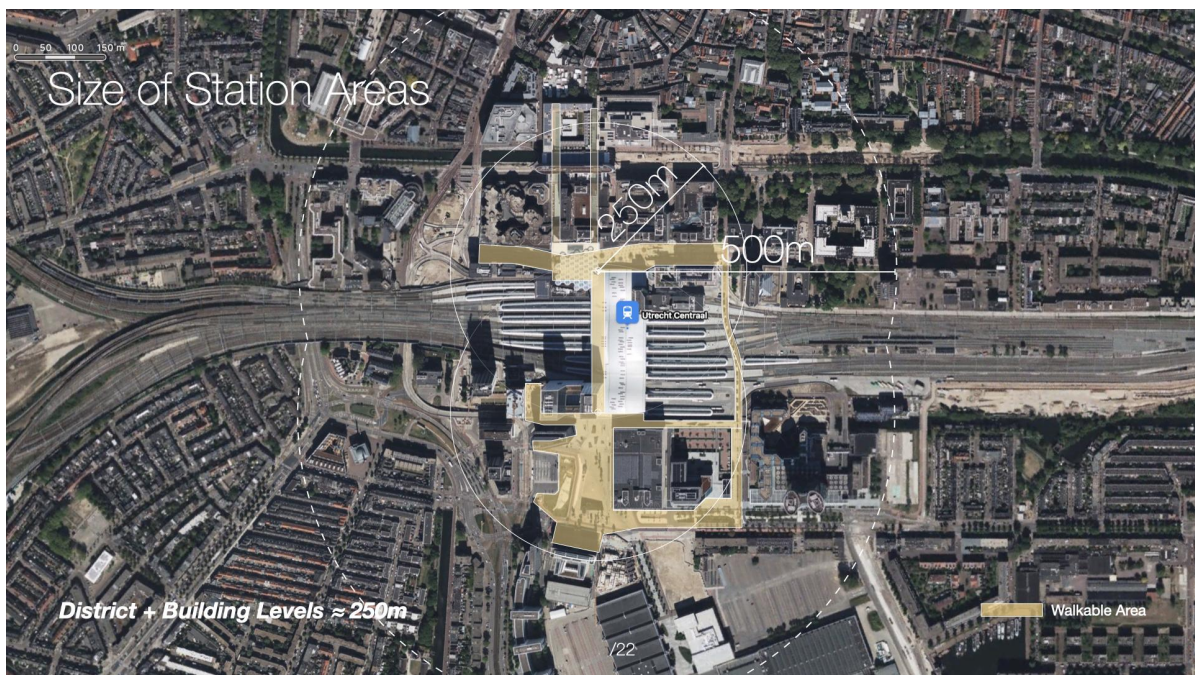


Figure 3: The Utrecht Central Station area - 250m vs 500m. (Image source: by authors)

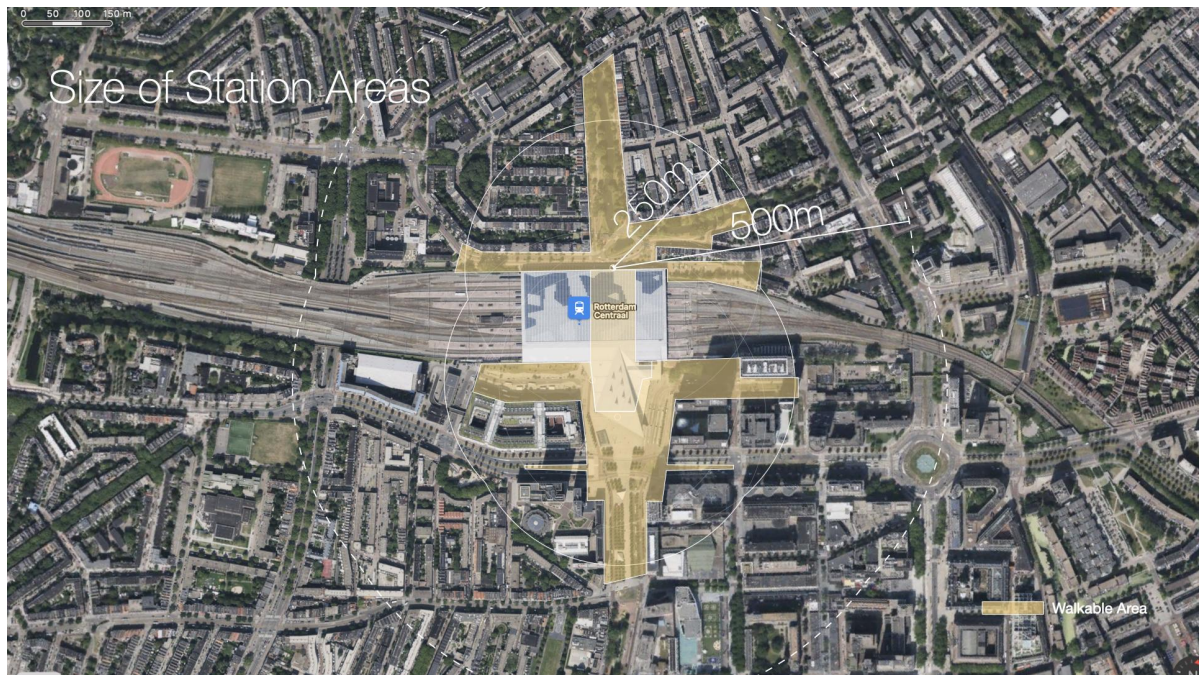


Figure 4: The Rotterdam Central Station area - 250m vs 500m. (Image source: by authors)



Figure 5: The Beijing West Railway Station area - 250m vs 500m. (Image source: by authors)

Literature and case selection

Enrich the meaning of 'fluctuation-responsive station areas' and establish the first theme for literature searching

Since overcrowding and underutilization are **very** particular issues that are barely addressed by any existing spatial design research, therefore if directly using 'railway station area' + 'fluctuation-responsive' as the keywords for searching, hardly any relevant papers **can** be found, and no heuristics **can** be gained. An enriched meaning of 'fluctuation-responsive station areas' is needed to serve as a basis for literature selection. This paper enriches the term's meaning by reinterpreting the identities of 'railway station areas' and the nature of 'overcrowding' and 'underutilization.'

The identities of railway stations: According to Bertolini and Spit (1998), 'railway stations' are 'nodes' of the transport networks and 'places' of the cities, so stations are fundamentally 'transport infrastructures' for transportation activities and 'public spaces' for other leisure activities.

'Station areas' include the 'station buildings' and the outdoor 'spaces,' therefore, 'open/urban/public space' and 'landscape' are relevant. 'Overcrowding' **can** be seen as a type of 'risk' to the station system, which **can** be addressed by 'resilience' as a quality of the system. 'Underutilization' is a lack of use of public spaces; therefore, 'flexible' use **can** be helpful. Both 'resilience' and 'flexible' mean a system that is capable of changing, and similar words **can** include 'dynamic,' 'adaptive,' 'robust,' 'moveable,' 'reconfigurable,' 'multi-functional (in the time dimension),' 'temporary,' 'kinetic', and so on. The combination of these concepts facilitates this paper in finding relevant literature.

Paper selection criteria

The papers that are used as the heuristics for research-by-design meet at least one of the following criteria: 1) The papers are related to design aspects, including design qualities, programming/functions, layouts, spaces, elements, and so on. 2) The papers deal with demands and users in station areas. 3) The papers' knowledge is preferably applicable on both building and outdoor spaces, otherwise too specific and not generalizable. In paper selection, **we** excluded plenty of papers that are purely about planning, or managerial solutions, which **can** hardly be transformed to spatial interventions.

Initial papers selected through each searching strategy and their relevance to the layouts and elements of design principles

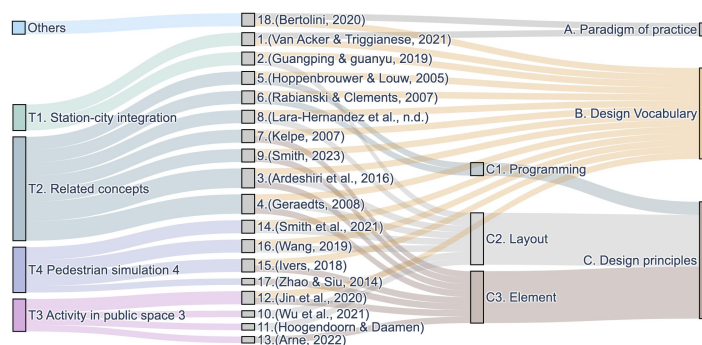


Figure 6: The literature selected and the relevance to the research findings

Initial station cases selected

Table 1: Cases selected

Stations	Size	Demand fluctuation	Performance for fluctuation	Nation
Beijing West RS	Large	Large	Poor	China
Beijing South RS	Large	Large	Poor	China
Shanghai RS	Large	Large	Good	China
Rotterdam Central Station	Large	Middle	Good	The Netherlands
Utrecht Central Station	Large	Middle	Good	The Netherlands
Bijlmer Arena RS	Middle	Large	Good	The Netherlands
Zandvoort Station	Small	Large	Good	The Netherlands

Data processing

Data processing

(Data processing for developing Design Principles)																					
		Design Objectives										Spatial or Managerial Dimensions									
		Problem-oriented (Macro-level)					User-based (Micro-level)					Spatial					Managerial-Spatial?				
		Has Overcoming phenomenon?	Has Interventions that (potentially) reduce Overcoming?	Has Interventions that (potentially) reduce Overcoming?	Has Interventions that (potentially) reduce Overcoming?	Add City Value?	Reduce Conflicts of Use between the Station and the City?	Related to Safety (Overcoming or Exposed to the Street)	Related to Speed	Related to Ease of Wayfinding	Related to Comfort of Facilities and Weather-Protected areas	Related to Mixed Experience	Related to Different Street (Scale)?	Related to Layout-Element?	Related to PathView?	Related to Programming?	Related to Access/Flow/Regulation?	Has Spatial relevance?	Feasible to Add Event?	Has Spatial relevance?	Other managerial aspects
Cases	1	Beijing West	Problem Statement	Problem Statement	Problem Statement	Design Principle 22, 23	o	Non-peak time Employees	Design Principle 19	o	Fast wayfinding in Complex, Congested Underground Station	o	o	o	o	o	o	o	o	o	o
	2	Shanghai	o	Design Principle 5	o	Design Principle 22, 23	Design Principle 22, 23	o	o	Design Principle 5 - Limitation	o	o	o	o	o	o	o	o	o	o	o
	3	Rotterdam	o	Design Principle 6, 19	o	Design Principle 6, 7, 16, 20, 21, 28	Design Principle 1, 7, 16, 20, 21, 23, 28	o	o	Different pattern, different model	o	o	o	o	o	o	o	o	o	o	o
	4	Utrecht	o	Design Principle 3	o	Design Principle 1, 7, 16, 20, 21, 23, 28	Design Principle 20, 21, 22, 23	o	o	Different pattern for different model	o	o	o	o	o	o	o	o	o	o	o
	5	Bilmer	o	Design Principle 5, 19	o	Design Principle 21, 22	o	Peak-time Overcoming	o	o	o	o	o	o	o	o	o	o	o	o	o
	6	Zandvoort	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	7	Baling South	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	8	Guangzhou	o	Design Principle 12	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	9	Shanghai South	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	10	Almere Amstel	o	Design Principle 18	o	Design Principle 18 - 19	o	o	o	Design Principle 15 - 16	o	o	o	o	o	o	o	o	o	o	o
Case Comparisons (Any differences or commonalities considering the different objectives/dimensions?)	1	Beijing West vs Beijing South	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	2	Beijing West vs Shanghai	o	Design Principle 5, 19	o	Design Principle 22, 23	Design Principle 22, 23	o	o	Design Principle 5	o	o	o	o	o	o	o	o	o	o	o
	3	Beijing West vs Rotterdam	o	Design Principle 6, 19	o	Design Principle 6, 7, 21, 22, 23, 28	Design Principle 20, 21, 22, 23	o	o	Design Principle 8	o	o	o	o	o	o	o	o	o	o	o
	4	Beijing West vs Utrecht	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	5	Beijing West vs Bilmer	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	6	Beijing West vs Zandvoort	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	7	Beijing South vs Shanghai	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	8	Beijing South vs Rotterdam	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	9	Beijing South vs Utrecht	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	10	Beijing South vs Bilmer	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	11	Beijing South vs Zandvoort	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	12	Shanghai vs Rotterdam	o	o	o	Design Principle 22, 23	Design Principle 22, 23	o	o	o	Security checks	o	o	o	o	o	o	o	o	o	o
	13	Shanghai vs Utrecht	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	14	Shanghai vs Bilmer	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	15	Shanghai vs Zandvoort	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	16	Rotterdam vs Utrecht	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	17	Rotterdam vs Bilmer	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	18	Rotterdam vs Zandvoort	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	19	Utrecht vs Bilmer	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	20	Utrecht vs Zandvoort	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	21	Bilmer vs Zandvoort	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
Literature	1	Van Acker & Tigghebeek, 2021	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	2	Guo & Li, 2019	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	3	Ardehian et al., 2019	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	4	Gewirth, 2008	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	5	Hoppenbrouwer & Looze, 2020	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	6	Palmer, 2007	o	o	o	Design Principle 16, 17	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	7	Smith, 2020	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	8	Hoppenbrouwer & Chavakis, 2020	o	Design Principle 16	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	9	Uut et al., 2020	o	Flow regulation	o	o	o	o	o	o	Flow regulation, with longer distance	o	o	o	o	o	o	o	o	o	o
	10	Wang, 2020	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	11	Smith et al., 2021	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o	o
	12	Wang, 2020	o	o	o	Design Principle 16, 17, 28	o	o	o	o	o	Design Principle 16, 17, 28	o	o	o	o	o	o	o	o	o
Legend:																					
Initial literature/cases/comparisons of cases		Abbreviations of stations:																			
Case comparisons that have not been examined		References:																			
Not relevant regarding this aspect																					
Examined but not relevant findings																					
Has relevant findings		Beijing West – Beijing West Railway Station																			
Has relevant and positive findings		Shanghai – Shanghai Railway Station																			
Has relevant and negative findings		Rotterdam – Rotterdam Central Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Rotterdam – Rotterdam Central Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			
Has relevant and positive findings		Bilmer – Bilmer Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and negative findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and positive findings		Utrecht – Utrecht Central Station																			
Has relevant and negative findings		Bilmer – Bilmer Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Guangzhou – Guangzhou Railway Station																			
Has relevant and positive findings		Zandvoort – Zandvoort Railway Station																			
Has relevant and negative findings		Utrecht – Utrecht Central Station																			

Figure 7: Examine relevance and compare cases

Internal reflections of design principles

(Internal evaluations of Design Principles)

Index	Names of Design Principles	Meet Design Objectives?									Have Spatial Relevance ?
		Problem-oriented (Macro-level)					User (Passenger)-based (Micro-level)				
		Reduce Overcrowding ?	Reduce Emptines ?	Add city value ?	Reduce conflicts of use between the station and the city?	Improve Safety (Overcrowding or Eyes-on-the-Street)	Improve Speed	Improve Ease (of Wayfinding)	Improve Comfort (of Facilities and Weather-Protected areas)	Improve (Visual) Experience	
1	Flexible use	√#	√#	√#	○	√#	#	#	√#	√+	√ (Programming)
2	Suitable general layouts of the station and city	√#	√#	√#	√#	√#	√#	√#	√#	√#	√ (Spatial Layout)
3	Set apart non-transport function	√+	○	○	√+	√#	○	√+	○	√#	√ (Spatial Layout)
4	Vibrate city environment by scattered mobility nodes	○	√+	√+	○	○	√-	√-	○	○	√ (Spatial Layout)
5	Increase accommodating capacity by scattered mobility nodes	√+	○	○	○	√+	√-	√-	○	○	√ (Spatial Layout)
6	Make events visible by aligning main paths and open spaces	○	√+	○	○	○	○	○	○	√+	√ (Spatial Layout; PathFlow)
7	Human-oriented spaces	○	√+	○	√+	○	○	○	○	√+	√ (Spatial Layout; Flow)
8	Connect with neighborhoods	√#	○	√#	√#	√+	○	○	○	○	√ (Layout)
9	Smooth level changes by landscape design (District level)	○	√+	○	○	○	○	○	○	○	√ (Spatial Element (District level))
10	Smooth level changes by landscape design (Building level)	○	√+	○	○	√+	○	○	○	○	√ (Spatial Element (Building level))
11	Adaptive redundant spaces with path regulation	√+	√+	○	○	√+	√+	○	○	○	√ (Spatial Layout; PathFlow)
12	Programming considering the time dimension	√+	√+	√+	○	○	○	○	○	○	√ (Programming)
13	Shortcuts or optimizing paths	○	○	○	√+	√-	√+	√-	○	√-	√ (Spatial Layout; PathFlow)
14	Flexible buildings	√+	√+	○	○	○	○	○	○	○	√ (Spatial Element (District level))
15	Flexible building components	√+	√+	○	○	○	○	○	√#	√#	√ (Spatial Element (Building level))
16	Add installations and facilities	○	√+	○	○	○	○	○	√+	√+	√ (Spatial Element)
17	Reconfigurable elements	○	√+	○	○	○	○	○	○	○	√ (Spatial Layout)
18	Redundant spaces	√+	√-	○	○	√+	√-	○	○	√-	√ (District level) Spatial Layout)
19	Set apart bottlenecks	√+	√-	○	○	√+	√-	○	○	○	√ (Building level) Spatial Layout)
20	City passage	○	√+	√+	○	○	√#	○	○	○	√ (Spatial Element; PathFlow)
21	Position city passages	√#	√+	√+	√#	○	○	○	○	○	√ (Spatial Layout; PathFlow)
22	Reduce barriers to ease flow (District level)	√#	√+	√+	√#	○	○	○	√+	○	√ (District level) Spatial Element; PathFlow)
23	Reduce barriers to ease flow (Building level)	√#	√+	√+	√#	○	○	○	√+	○	√ (Building level) Spatial Element; PathFlow)
24	Stairs as stages or seats	○	√+	○	○	○	○	○	○	√+	√ (Spatial Element)
25	City axes	√#	√+	√+	○	○	○	√+	√+	○	√ (Spatial Layout; PathFlow)
26	Reconfigurable spaces	○	√+	○	○	○	○	○	○	√+	√ (Spatial Element)

Legend:

○	No relevance
√	Has relevance
√+	Has relevance, with Positive effects
√-	Has relevance, with Negative effects
√#	Has relevance, and the effects depend on the contexts
#	Uncertain

Figure 8: The table of internal evaluation of design principles

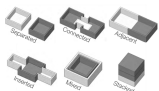
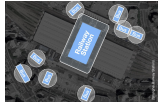
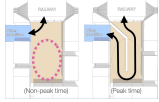
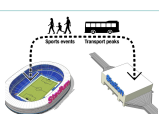
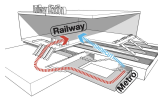
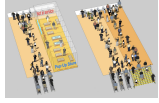
Results and interpretations

Sources of heuristics for each design principle

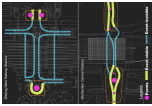
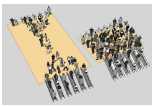
Table 2: Sources of heuristics for each design principle

Diagrams	Minor design principles	Cases	Literature	Analytical visualization	Reflections from experts
	1) Flexible use	Paris Beaches	Hoppenbrouwer - and Louw 2005	-	-

Continued on next page

	2) Suitable general layouts of the station and city	-	Qi and Lu 2019, Shikata et al. 2013	Apply different layouts	-
	3) Set apart non-transport function	Utrecht RS current vs former; -	-	-	-
	4) Enliven city environment by scattered mobility nodes	-	-	Transformed from principle 5 to a larger scale	-
	5) Increase accommodating capacity by scattered mobility sites	Beijing West RS vs Shanghai RS	-	-	-
	6) Make events visible by aligning main paths and open spaces	Beijing West RS; Rotterdam RS current vs former	-	Virtually add events to see the effects	-
	7) Human-oriented spaces	Rotterdam RS current vs former	-	Virtually add events to see the effects	-
	8) Connect with neighborhoods	Beijing West RS; Rotterdam RS; Utrecht RS	Peters and Tolkoﬀ 2016	-	Mind transport and non-transport interferences
	9) Smooth level changes by landscape design (District level)	Jiaxing RS	-	Transformed from principle 10 to a larger scale	-
	10) Smooth level changes by landscape design (Building level)	Arnhem RS	-	-	Costly, harder wayfinding
	11) Layouts that support path reconfigurations	Shanghai RS; (Path regulations in Airports; Shopping malls)	-	Transformed from non-station case experience	Path regulation being intrusive, managerial efforts
	12) Programming considering the time dimension	Guangzhou; TOD mixed-use practice	Calthorpe and Poticha 1993	Transformed from higher-frequency events	-
	13) Shortcuts or optimizing paths	Beijing West RS	-	Design shortcuts as a new solution	Extra cost
	14) Flexible buildings	The Shed; Stadium 974	Ivers 2018, Smith et al. 2021	-	Costly
	15) Flexible building components	-	-	Transformed from principle 14 to a smaller scale	-

Continued on next page

	16) Add installations and facilities	Antwerpen RS	Ivers 2018, Kelppe 2007	Transformed from non-station cases	Installations' quality matters
	17) Reconfigurable elements	-	Ivers 2018	Transformed from non-station cases	-
	18) redundant spaces	Allianz Arena	-	Transformed from non-station cases	-
	19) Setting apart bottlenecks	-	Hoogendoorn and Daamen 2004	-	-
	20) City passages	Rotterdam RS; Utrecht RS; Bijlmer RS	-	-	-
	21) Position city passages	Rotterdam; Utrecht; Amsterdam	-	Apply this as a context for principle 20	-
	22) Reduce barriers to ease flow (District level)	Beijing West RS; Rotterdam RS; Utrecht RS	-	-	-
	23) Reduce barriers to ease flow (Building level)	Beijing West RS	-	Transformed from principle 22 to a smaller scale	-
	24) Stairs as stages or seats	Utrecht RS; Cologne RS; Antwerp RS	-	-	-
	25) City axes	Rotterdam RS; Utrecht RS	-	-	-
	26) Reconfigurable spaces	-	Ivers 2018	Transformed from non-station cases	Managerial efforts

For the simplicity of the final report, the design principles are clustered into six major principles, each containing multiple minor principles. [Table 6.2](#) provides a brief overview of the major design principles and the main heuristics they are based on. These six principles address a range of crucial design aspects, including spaces, elements, layouts, and functions.

Relate design principles to an event typology

Events are an important aspect of this research, as they are the causes of fluctuations in station areas and are a solution to address underutilization. As part of the exploration of design solutions, we investigated the relationship between event types and design principles.

In this research, we adopted an event typology developed by Smith et al. (2021). This typology is framed based on access and mobility, including nine types of events: E1) large indoor arenas, E2) concerts and screenings, E3) pleasure gardens, E4) parades and street races, E5) roadshows, E6) markets and fairs, E7) mass gatherings, E8) micro-performances, and E9) marches and street parties. Two events - E4 and E9, primarily happen on streets, which are irrelevant for most spaces in railway stations, so we excluded (✕) them (Figure 9).

The relationships between event types and design principles, are explained as follows (Note: 'E' stands for Event types, and 'D' stands for Design principles):

- Large open-air events with concentrated distributions, including E2, E3, and E7, need large coherent outdoor spaces and should not block city passages. Therefore, these events have relevance to design principles (DPs) (11) Adaptive redundant spaces with path regulation, (7) Human-oriented spaces, and (21) City passages' positioning. (Links: E2-D7, E3-D7, E6-D7; E7-D7; E2-D11, E3-D11, E6-D11, E7-D11; E2-D21, E3-D21, E6-D21, E7-D21)
- For large events with performances - E2, (24) Stairs as stages or seats can be used. (Link: E2-D24)
- Small spaces are common in a scattered layout (D4), and small events - E8 can be held in these small spaces. (15) Flexible building components can also be the small spaces to hold small events. (Links: E8-D4, E8-D15)
- For events that require user participation and are open-access, including E7, E5, E6, and E8, it is necessary to (6) Make events visible by aligning main paths and open spaces. (Links: E7-D6, E5-D6, E6-D6, E8-D6)
- Small performances and markets can help promote a vibrant environment, and it is a prerequisite to (23) Reduce barriers to ease flow at the building level. (Links: E6-D23, E8-D23)
- Large indoor arenas can be Flexible buildings (14). Flexible and Shared use between arenas and stations can be realized if the station areas are Programmed considering the time dimension. The arenas should be located in Suitable general layouts of the station and city (2). (Links: E1-D2, E1-D14, E1-D12)
- In events E3 and E6, it is necessary to (16) Add installations and facilities, which can be (17) Reconfigurable elements. (Links: E3-D11, E6-D11; E3-D16)
- For all possible events happening in railway stations, the event spaces should support flexible use and be configurable, and pedestrians should be able to walk smoothly to the event spots. Therefore, several design principles are relevant, including (22) Reduce barriers to ease flow at the district level, (25) City axes, (1) Flexible use, (26) Reconfigurable spaces. (Links: All-D25, All-D22, All-D26, All-D1)

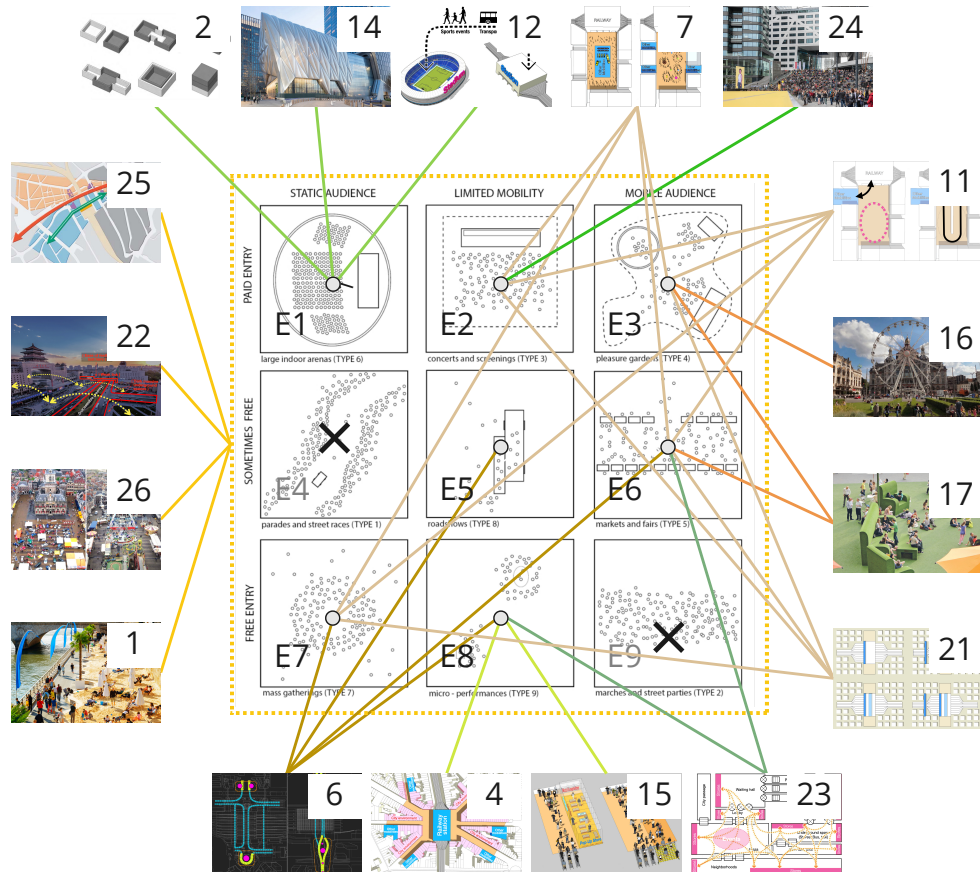


Figure 9: The design principles related to the event typology (Image source: Drawn by authors; The event typology diagram is reused from (Smith et al., 2021);